

ASM-HEMT Model for GaN RF and Power Electronic Applications

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MA Long 馬龍

EEsof EDA

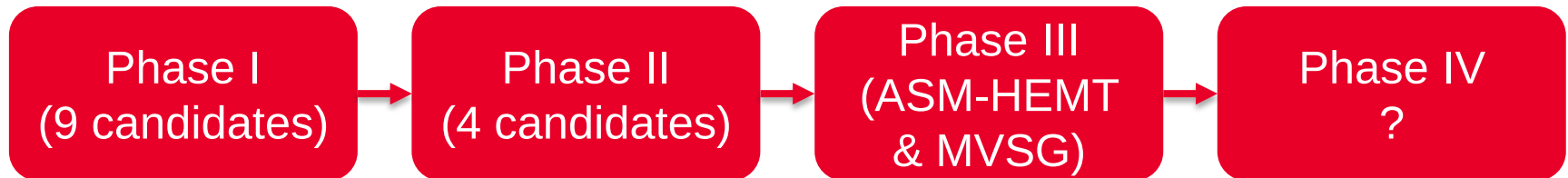
long_ma@keysight.com

Agenda

- **ASM-HEMT Model Overview**
- Parameter Extraction
 - DC Model
 - Field Plate Capacitance Model
 - RF Small Signal Model
- Summary

GaN HEMT Model Standardization Effort in CMC

- The first CMC standard model for III-V semiconductor
- Requirements on the model
 - Physics-based
 - Gummel-symmetry
 - Applicable to both RF and PE applications
- Standardization process



ASM-HEMT Model Overview

$$n_s = DV_{th} \left\{ \ln \left[\exp \left(\frac{E_f - E_0}{V_{th}} \right) + 1 \right] + \ln \left[\exp \left(\frac{E_f - E_1}{V_{th}} \right) + 1 \right] \right\}$$

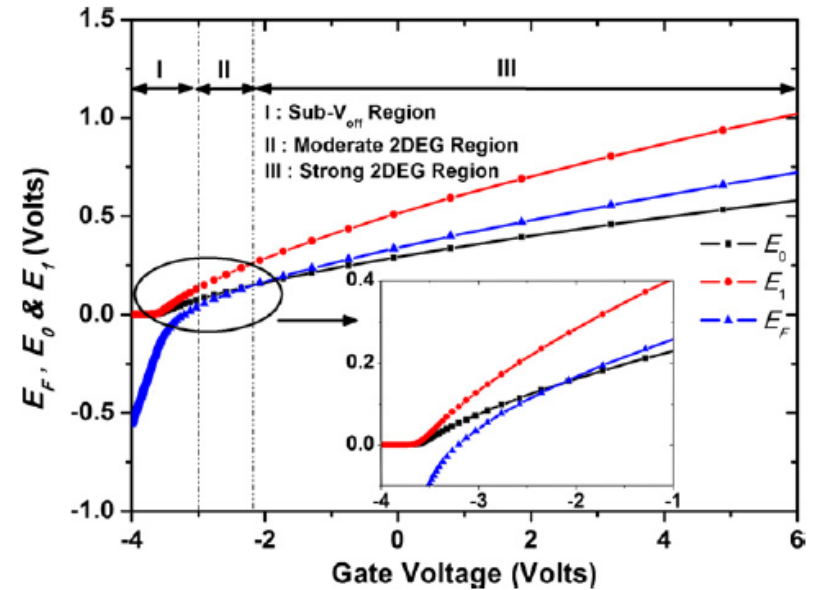
$$E_0 = \gamma_0 n_s^{2/3} \quad E_1 = \gamma_1 n_s^{2/3}$$

$$n_s = \frac{\varepsilon}{qd} (V_{go} - E_f - V_x)$$

Transcendental

$$E_{f,unified} = V_{go} - \frac{2V_{th} \ln \left(1 + e^{\frac{V_{go}}{2V_{th}}} \right)}{\frac{1}{H(V_{go}, p)} + \frac{C_g}{qD} e^{-\frac{V_{go}}{2V_{th}}}}$$

$$\psi = E_f + V_x$$



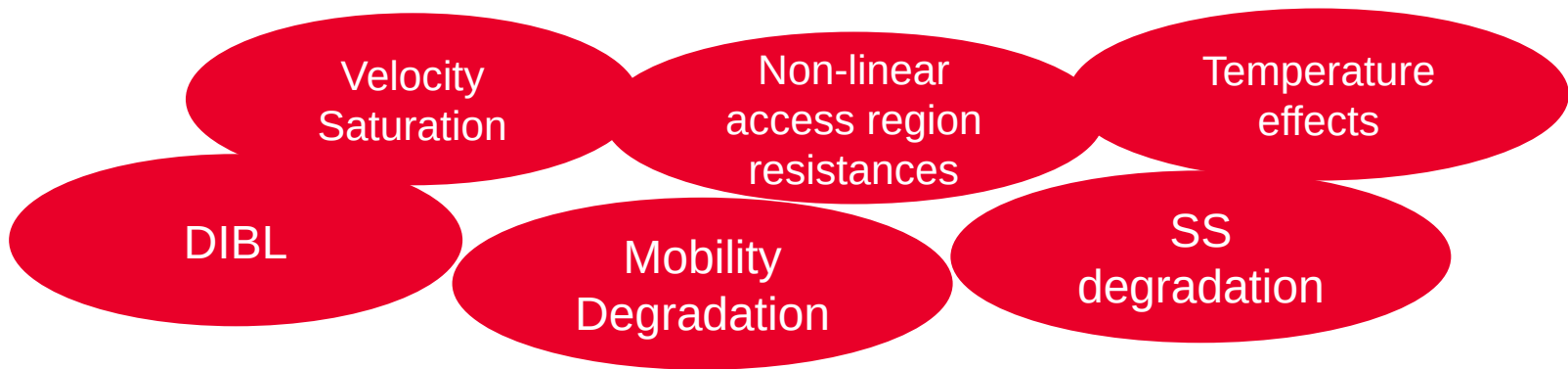
Drain Current Model

$$I_d(x) = \mu W N_f Q_{ch} \frac{d\psi(x)}{dx} + \mu W N_f V_{th} \frac{dQ_{ch}}{dx} \quad Q_{ch} = -C_g(V_{go} - \psi(x))$$

$$I_d = \frac{W}{L} \mu C_g N_f \left[V_{go} - \left(\frac{\psi_s + \psi_d}{2} \right) + V_{th} \right] (\psi_d - \psi_s)$$

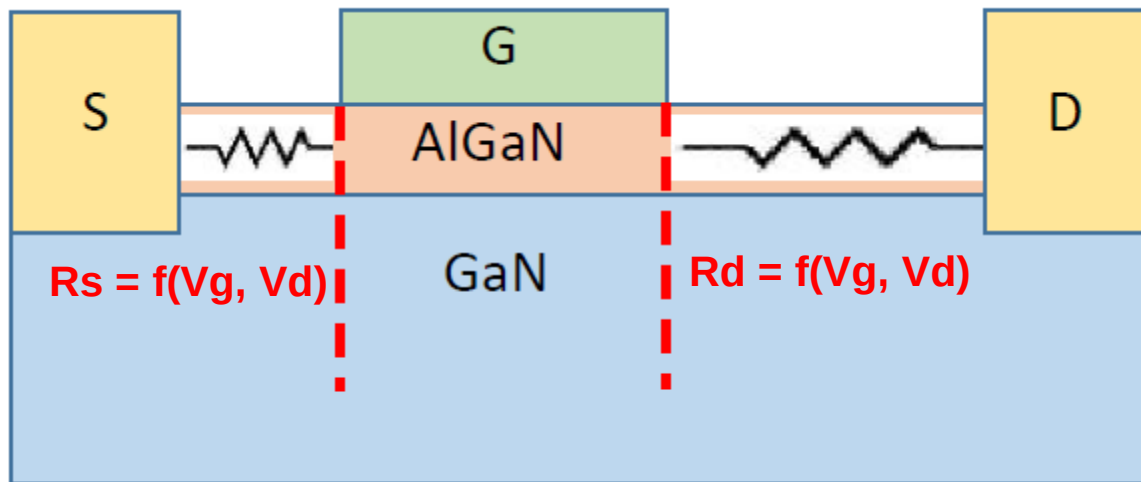
$$I_d = \frac{\mu_{eff}}{\sqrt{1 + \theta_{sat}^2 \psi_{ds}^2}} \frac{W}{L} C_g N_f \left[V_{go} - \left(\frac{\psi_s + \psi_d}{2} \right) + V_{th} \right] \psi_{ds} (1 + \lambda V_{ds})$$

Incorporating realistic device effects



Access Region Resistance Model

- Non-linear model which accounts for velocity saturation in access region
- It includes bias dependent access region resistance, source contact resistance and resistance due to traps.



Intrinsic Charges Model

- Surface-Potential Based Charge Expressions
- Ward-Dutton's partitioning method is applied for source and drain charges

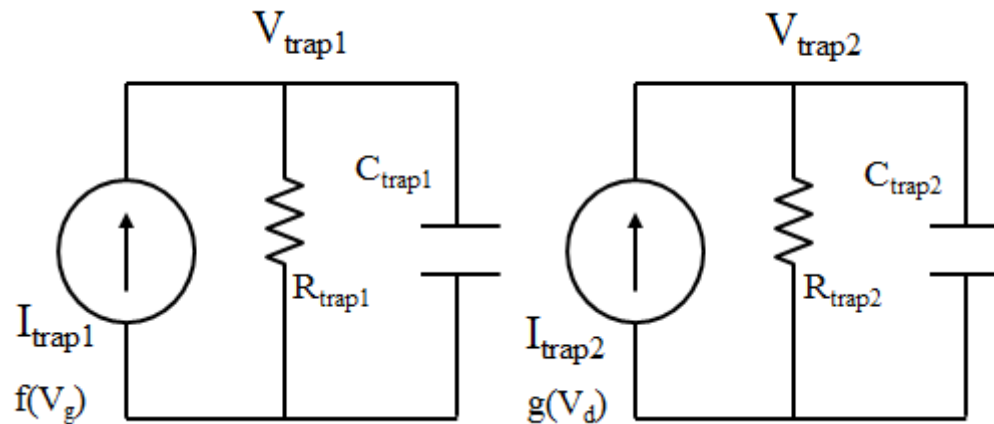
$$Q_{gi} = - \int_0^L W N_f C_g (V_{go} - \psi(x)) dx$$

$$Q_{di} = \int_0^L (x/L) W N_f C_g (V_{go} - \psi(x)) dx$$

$$Q_{si} = -Q_{gi} - Q_{di}$$

Trap Model

- Two RC networks for gate and drain lag
- Trap voltages V_{trap1} and V_{trap2} are fed back into expressions for Access region resistances, Cutoff Voltage and Subthreshold Slope



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List of Parameters I

Core Model Parameters

<code>parameter real VOFF = -2.0</code>	<code>from [-100.0:5];</code>	<code>//Cut-off voltage</code>
<code>parameter real U0 = 170.0e-3</code>	<code>from [0.0:inf);</code>	<code>//Low field mobility</code>
<code>parameter real UA = 0.0e-9</code>	<code>from [0.0:inf);</code>	<code>//Mobility Degradation coefficient first order</code>
<code>parameter real UB = 0.0e-18</code>	<code>from [0.0:inf);</code>	<code>//Mobility Degradation coefficient second order</code>
<code>parameter real VSAT = 1.9e5</code>	<code>from [1.0e3:inf);</code>	<code>//Saturation Velocity</code>
<code>parameter real DELTA = 2.0</code>	<code>from [2.0:inf);</code>	<code>//Exponent for <u>Vdeff</u></code>
<code>parameter real AT = 0.0</code>	<code>from (-inf:inf);</code>	<code>//Temperature Dependence for saturation velocity</code>
<code>parameter real UTE = -0.5</code>	<code>from [-10.0:0];</code>	<code>//Temperature dependence of mobility</code>
<code>parameter real LAMBDA = 0.0</code>	<code>from [0.0:inf);</code>	<code>//Channel Length Modulation Coefficient</code>
<code>parameter real ETA0 = 1.0e-9</code>	<code>from [0.0:inf);</code>	<code>//DIBL Parameter</code>
<code>parameter real VDSSCALE = 5.0</code>	<code>from (0.0:inf);</code>	<code>//DIBL Scaling VDS</code>
<code>parameter real KT1 = 0.0e-3</code>	<code>from (-inf:inf);</code>	<code>//Temperature Dependence for <u>Voff</u></code>
<code>parameter real THESAT = 1.0</code>	<code>from [1.0:inf);</code>	<code>//Velocity Saturation Parameter</code>
<code>parameter real NFACTOR = 0.5</code>	<code>from [0.0:inf);</code>	<code>//Sub-VOFF Slope parameters</code>
<code>parameter real CDSCD = 1.0e-3</code>	<code>from [0.0:inf);</code>	<code>//Sub-VOFF Slope Change due to Drain Voltage</code>
<code>parameter real IMIN = 1.0e-15</code>	<code>from [0.0:inf);</code>	<code>//Minimum Drain Current</code>

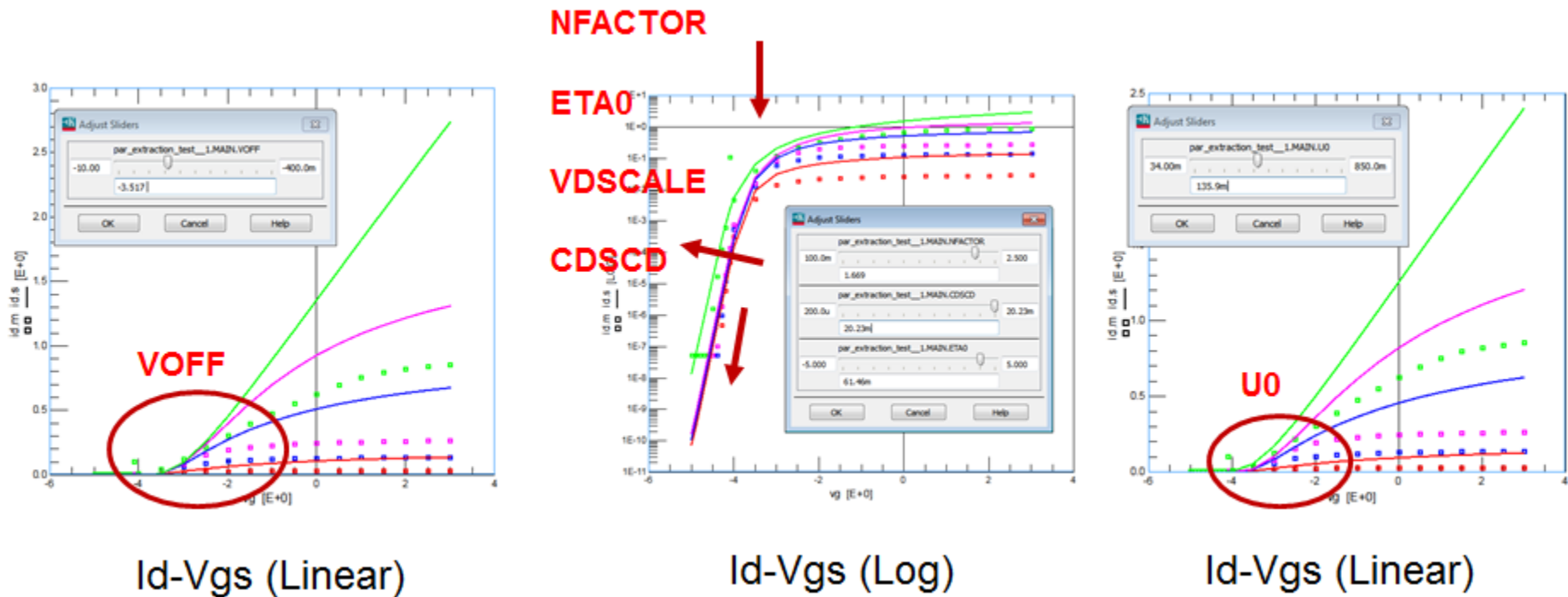
List of Parameters II

Access Region and Temperature Parameters

<code>parameter integer RDSMOD = 1</code>	<code>from [0:1];</code>	<code>//Switch for external source and drain resistances</code>
<code>parameter real VSATACCS = 50.0e3</code>	<code>from [0.0:inf);</code>	<code>//Saturation Velocity for access region: Source Side</code>
<code>parameter real NS0ACCS = 5.0e17</code>	<code>from [1.0e5:inf);</code>	<code>//2-DEG Charge Density in per square meter in Source access region</code>
<code>parameter real NS0ACCD = 5.0e17</code>	<code>from [1.0e5:inf);</code>	<code>//2-DEG Charge Density in per square meter in Drain access region</code>
<code>parameter real KOACCS = 0.0</code>	<code>from [0.0:inf);</code>	<code>//V_{gs} dependence parameter of source side access region 2-DEG charge density</code>
<code>parameter real KOACCD = 0.0</code>	<code>from [0.0:inf);</code>	<code>//V_{gs} dependence parameter of drain side access region 2-DEG charge density</code>
<code>parameter real UCACCS = 155e-3</code>	<code>from [0.0:inf);</code>	<code>//Access region mobility source-side</code>
<code>parameter real UCACCD = 155e-3</code>	<code>from [0.0:inf);</code>	<code>//Access region mobility drain-side</code>
<code>parameter real MEXPACCS = 2.0</code>	<code>from [0.0:inf);</code>	<code>//Exponent for access region resistance model</code>
<code>parameter real MEXPACCD = 2.0</code>	<code>from [0.0:inf);</code>	<code>//Exponent for access region resistance model</code>
<code>parameter real LSG = 1.0e-6</code>	<code>from [0.0:inf);</code>	<code>//Length of Source-Gate Access Region</code>
<code>parameter real LDG = 1.0e-6</code>	<code>from [0.0:inf);</code>	<code>//Length of Drain-Gate Access Region or Length of drain side access region</code>
<code>parameter real RSC = 1.0e-4</code>	<code>from [0.0:inf);</code>	<code>//Source Contact Resistance</code>
<code>parameter real RDC = 1.0e-4</code>	<code>from [0.0:inf);</code>	<code>//Drain Contact Resistance</code>
<code>parameter real KNS0 = 0.0</code>	<code>from [0.0:inf);</code>	<code>//Temperature Dependence for 2-DEG charge density at access region</code>
<code>parameter real ATS = 0.0</code>	<code>from (-inf:inf);</code>	<code>//Temperature Dependence for saturation velocity at access region</code>
<code>parameter real UT'S = 0.0</code>	<code>from (-inf:inf);</code>	<code>//Temperature dependence of mobility at access region: Source Side</code>
<code>parameter real UT'D = 0.0</code>	<code>from (-inf:inf);</code>	<code>//Temperature dependence of mobility at access region: Drain Side</code>
<code>parameter real KRSC = 0.0</code>	<code>from [0.0:inf);</code>	<code>//Temperature dependence of Source Contact Resistance</code>
<code>parameter real KRDC = 0.0</code>	<code>from [0.0:inf);</code>	<code>//Temperature dependence of Drain Contact Resistance</code>
<code>parameter real RTH0 = 5.0</code>	<code>from [0.0:inf);</code>	<code>//Thermal Resistance</code>
<code>parameter real CTH0 = 1.0e-9</code>	<code>from [0.0:inf);</code>	<code>//Thermal Capacitance</code>

Parameter Extraction I

Id-Vg Linear and Log Scale (Linear Vd Condition)



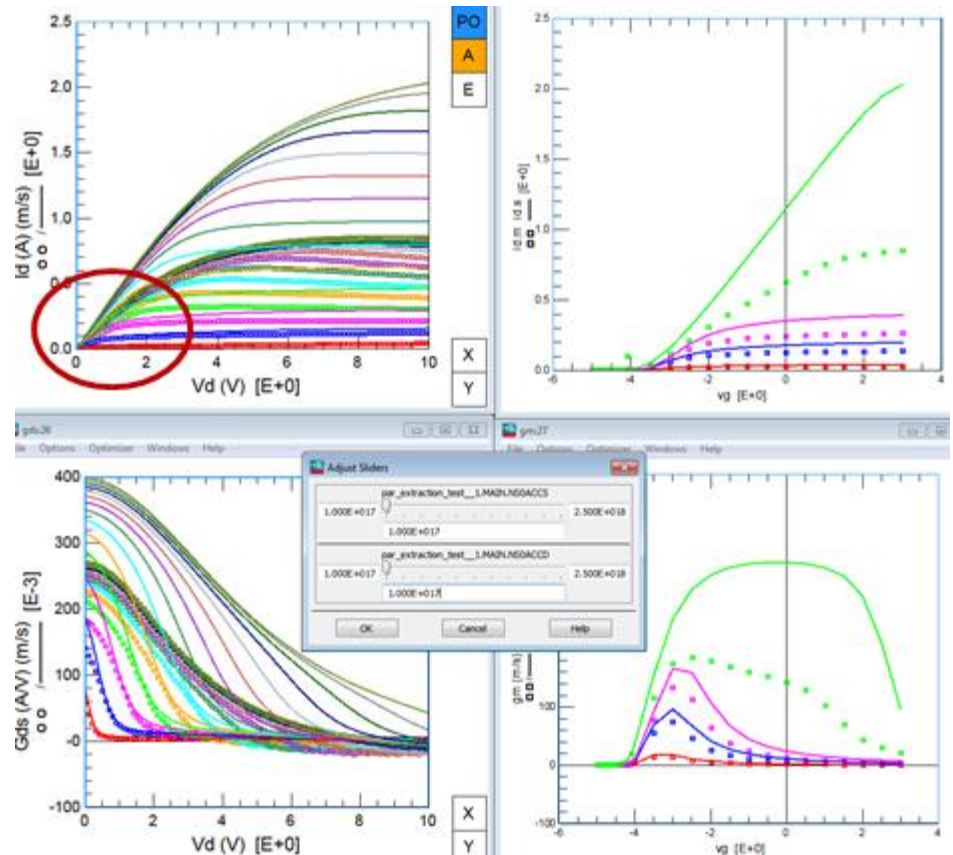
Parameter Extraction II

**I_d - V_{gs} , g_m - V_{gs} and I_d - V_{ds} ,
 g_{ds} - V_{ds}**

Adjust R_{on} using

NS0ACCS

NS0ACCD

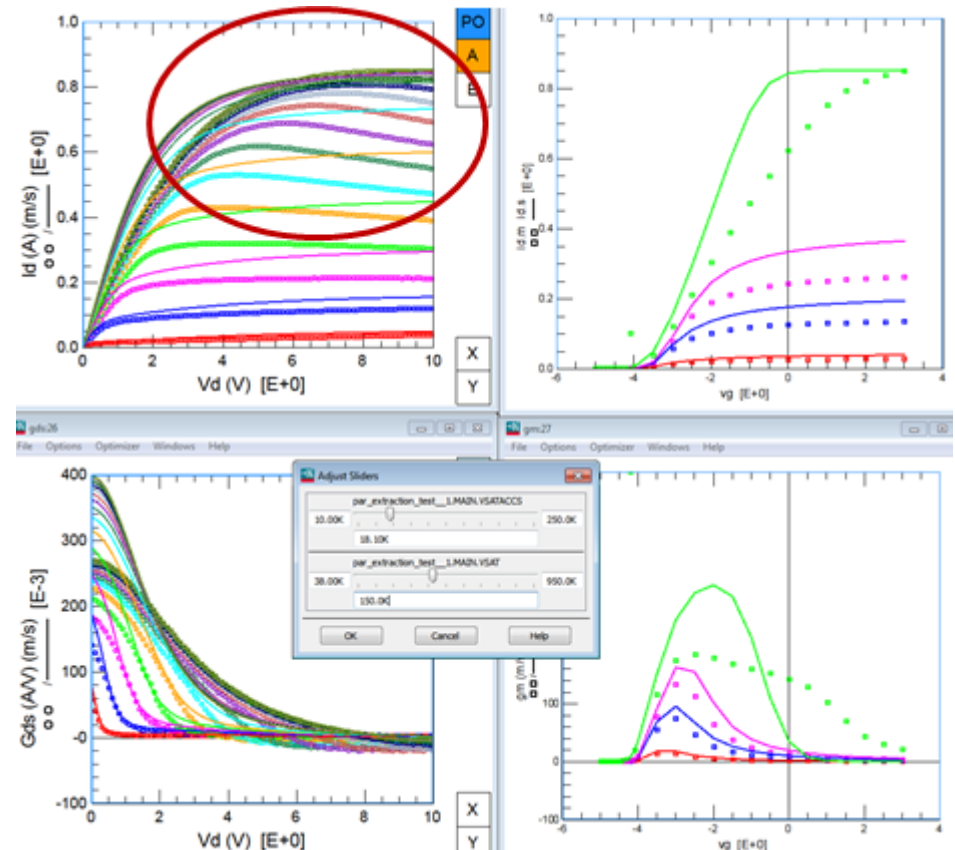


Parameter Extraction III

**I_d - V_{gs} , g_m - V_{gs} and I_d - V_{ds} ,
 g_{ds} - V_{ds}**

Fit Saturation Current by
adjusting **$VSATACCS$**

$VSAT$ should remain more or
less the same since it is a
material dependent parameter
and should not vary much for
different GaN Technologies



Parameter Extraction IV

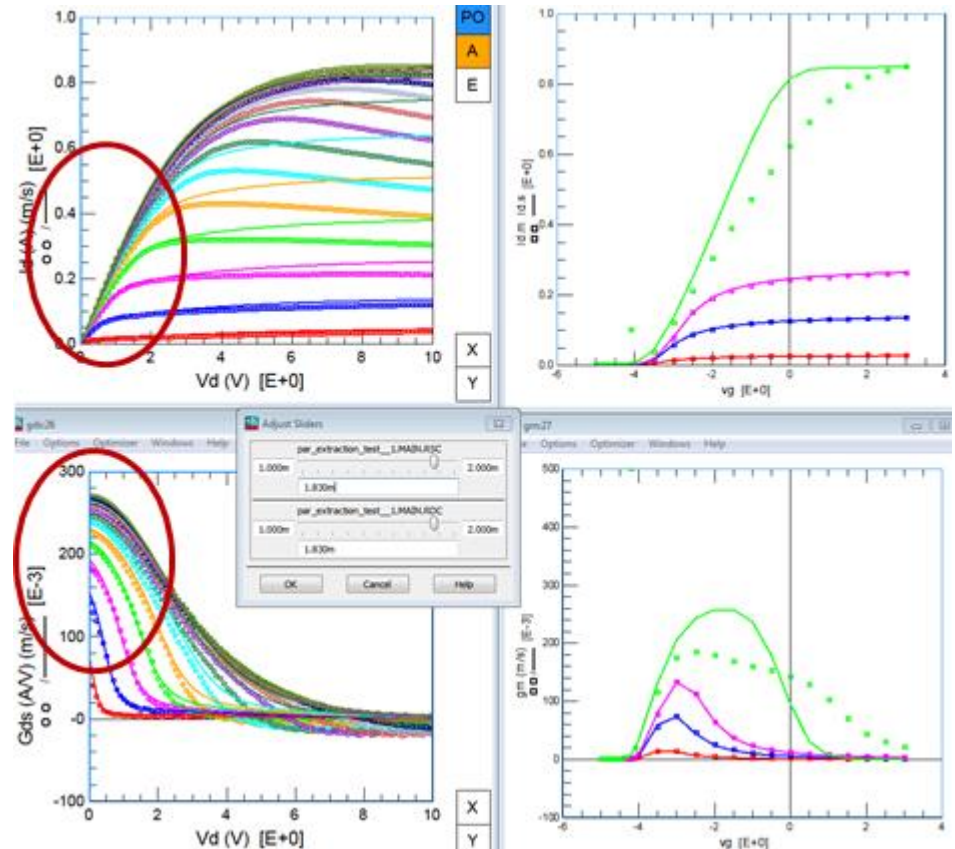
**I_d - V_{gs} , g_m - V_{gs} and I_d - V_{ds} ,
 g_{ds} - V_{ds}**

Fine tune Contact Resistance
parameters

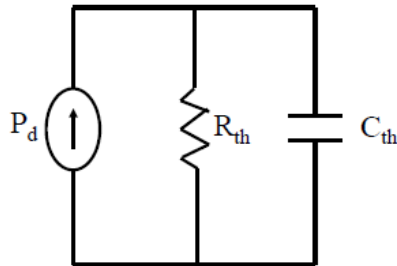
RDC

RSC

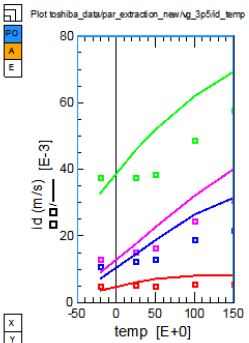
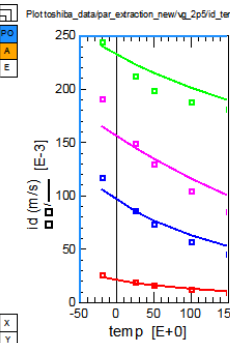
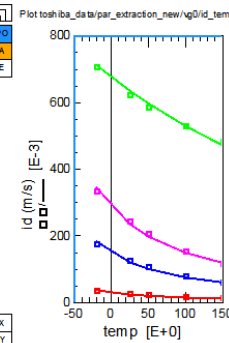
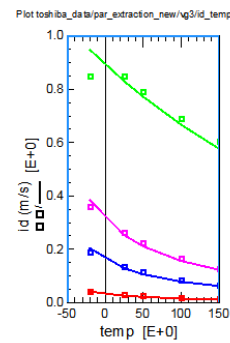
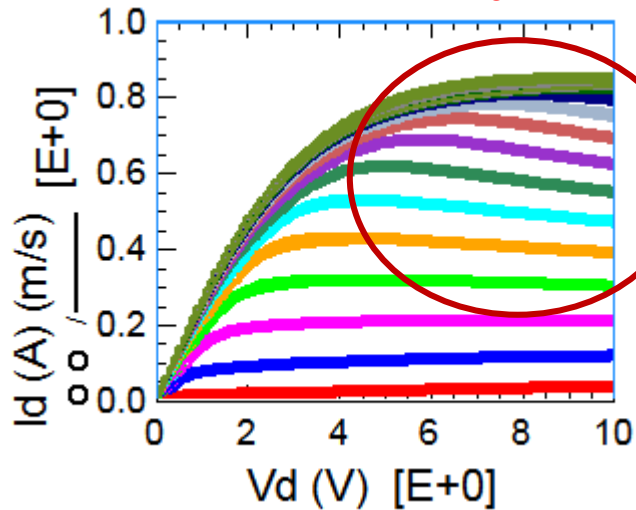
To further fit the R_{on}



Self Heating and Temperature Related Parameters



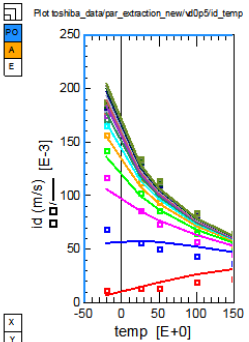
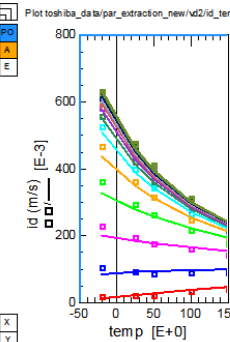
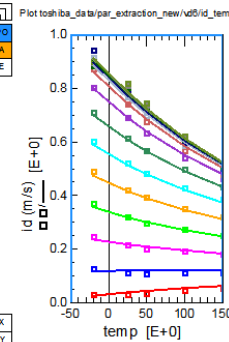
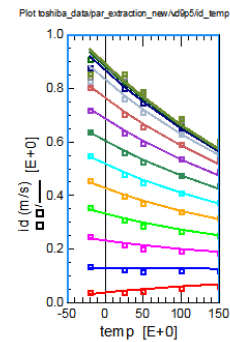
RTH0



KT1

AT

ATS

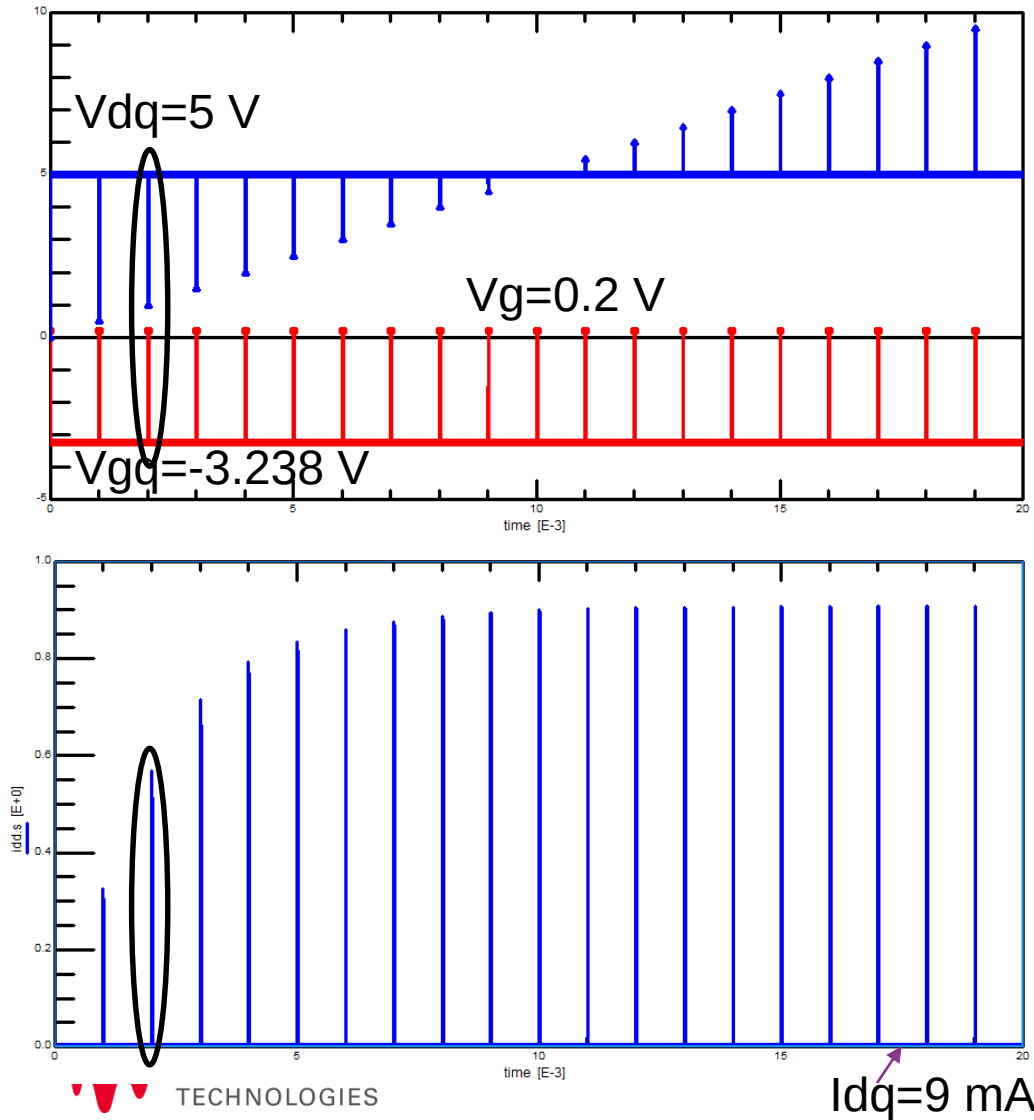


KRSC

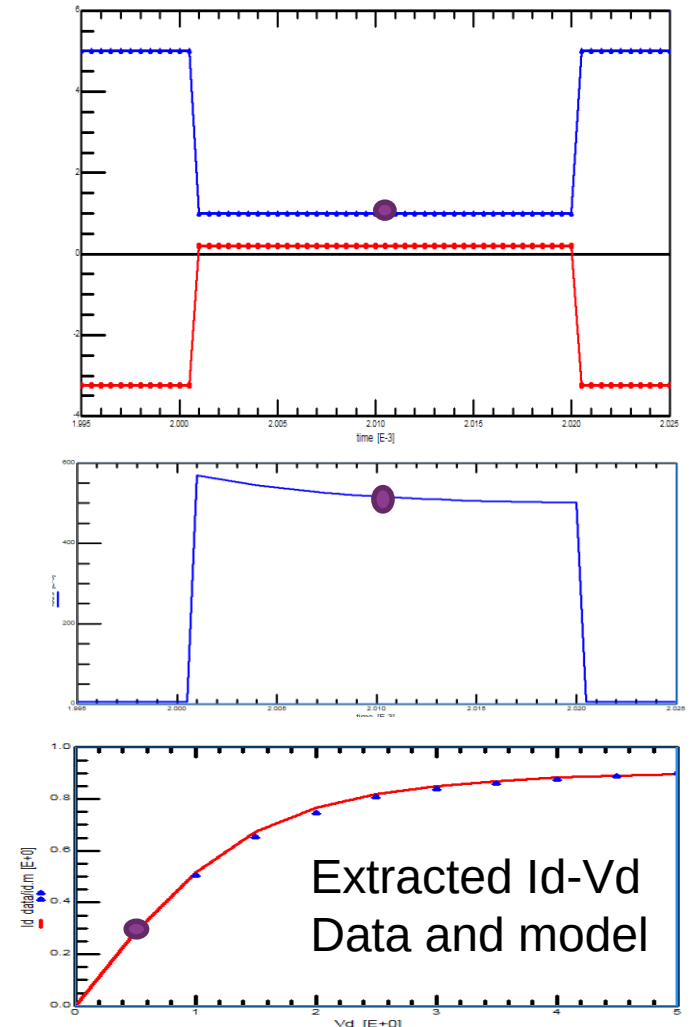
KRDC

UTE

Pulsed IV Setup

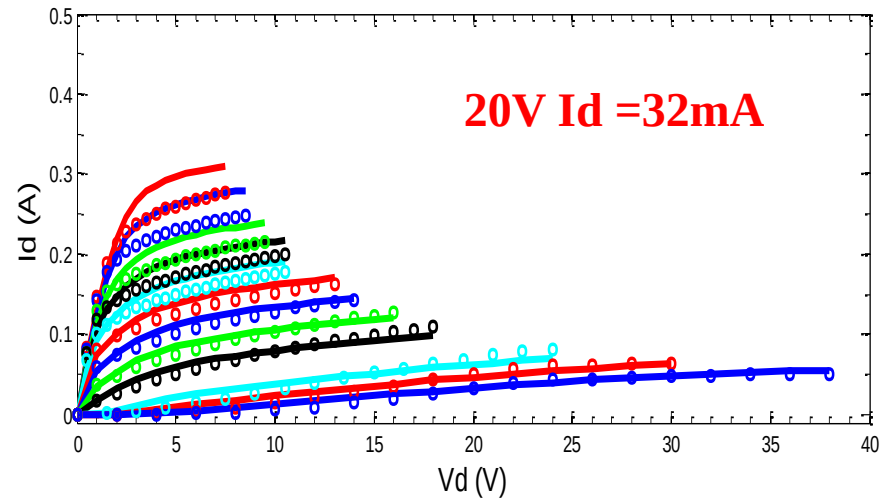
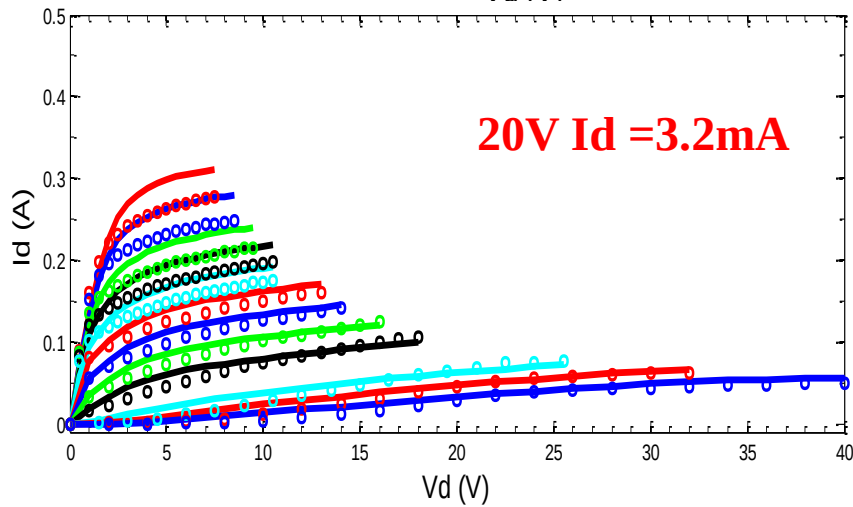
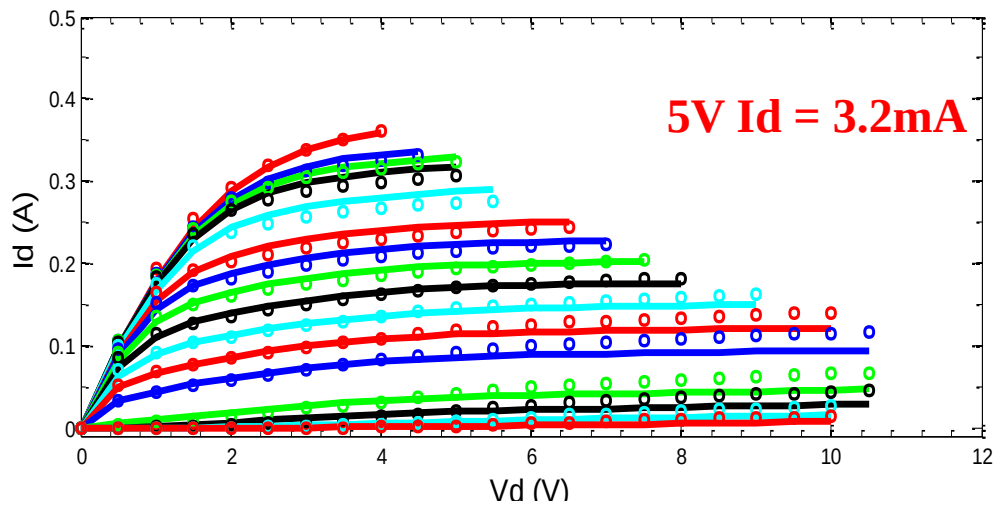


Zoomed Picture



Extracted I_d - V_d
Data and model

Pulsed IV Model Results



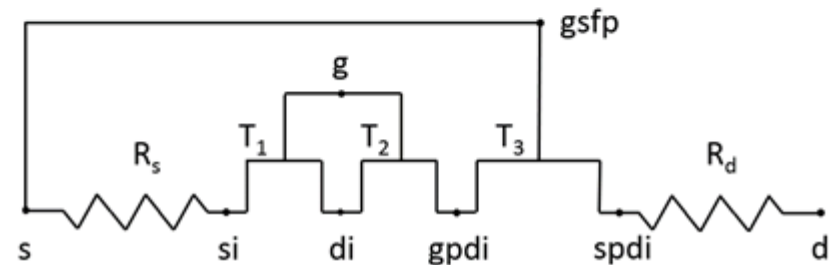
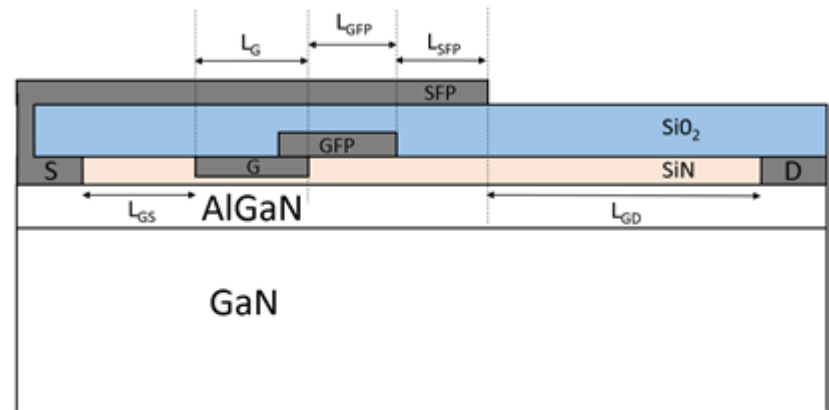
Field Plate Capacitance Model

Terminal Capacitances

- $C_{iss} = C_{gs} + C_{gd}$
- $C_{rss} = C_{gd}$
- $C_{oss} = C_{ds} + C_{gd}$

FP Modeled as an intrinsic HEMT

- Each HEMT governed by ψ calculation



Field Plate MOD Parameter Extraction I

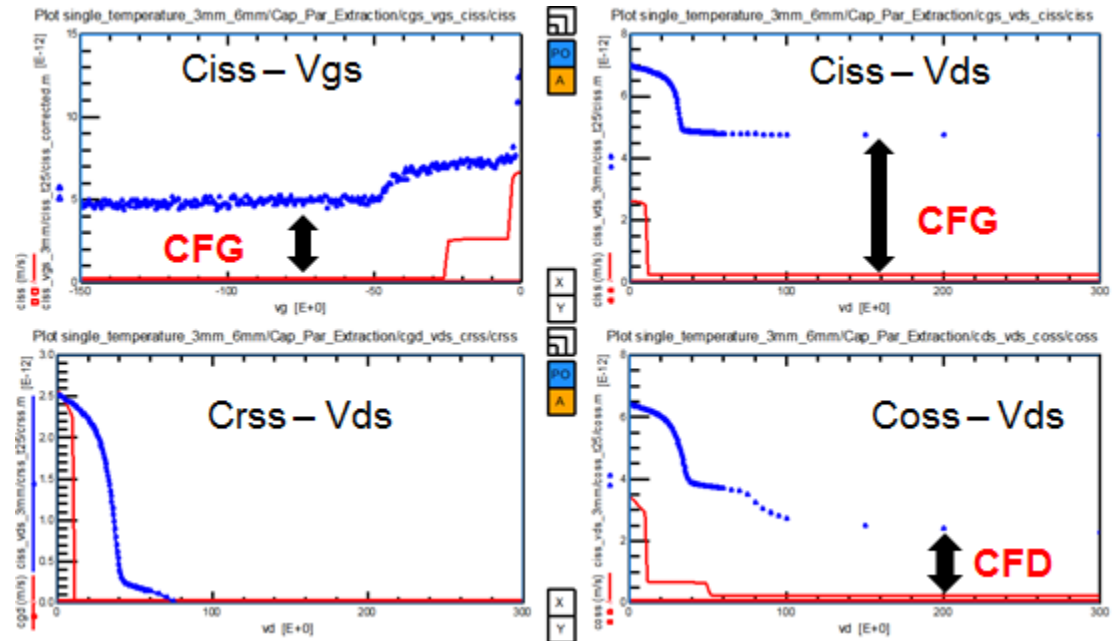
Ciss – Vgs

- $-150\text{V} < V_{gs} < 0\text{V}$
- $V_{ds} = 0$

Crss, Ciss, Coss versus Vds

- $0\text{V} < V_{ds} < 300\text{V}$
- $V_{gs} = -15\text{V}$

$$\text{GFPMOD} = \text{SFPMOD} = 1$$

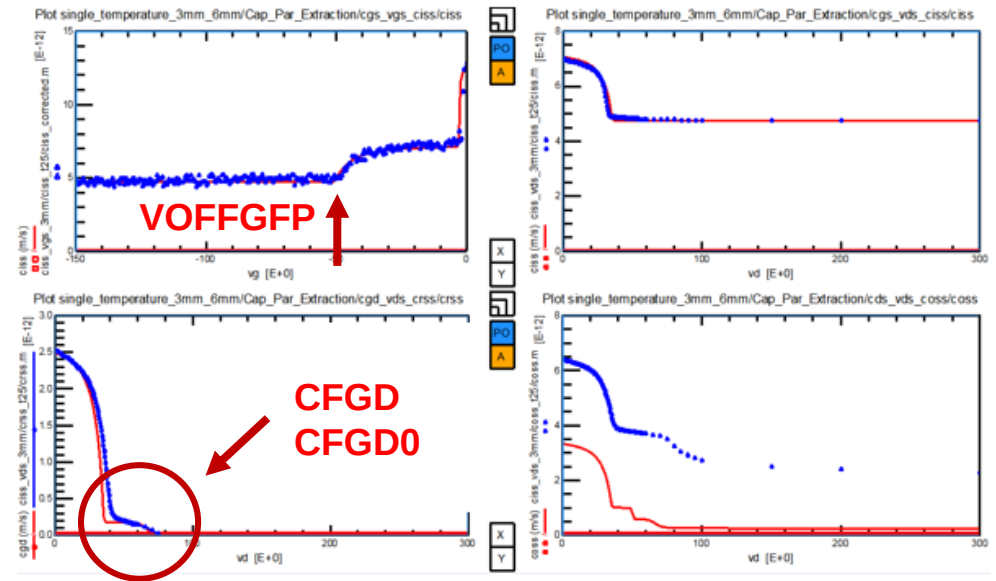


Field Plate MOD Parameter Extraction II

VOFFGFP – sets the rise of the hump

ADOS – smoothens the hump

CFGD } Gate FP - Drain
CFGD0 } Fringing capacitance



Field Plate MOD Parameter Extraction III

Optimize the following

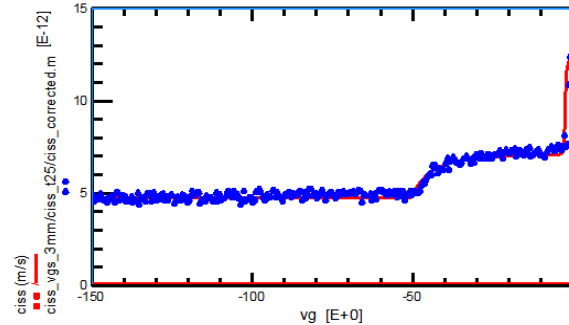
AJ

CJ0

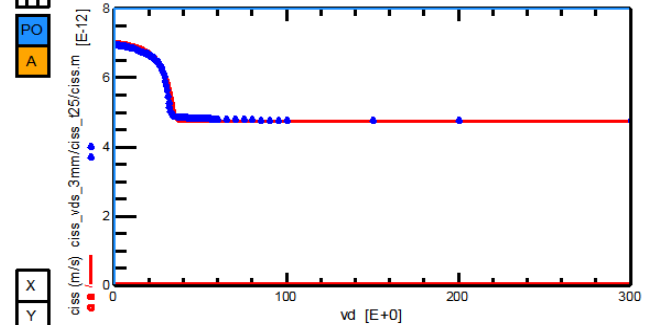
MZ

VBI

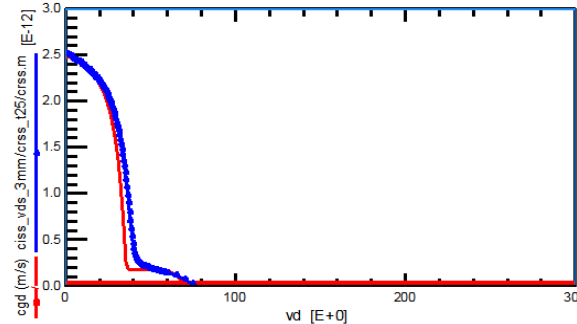
Plot single_temperature_3mm_6mm/Cap_Par_Extraction/cgs_vgs_ciss/ciss



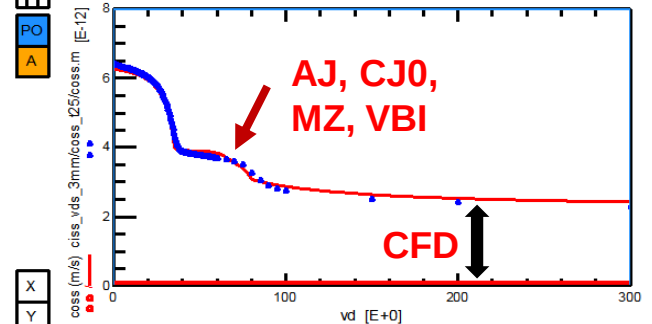
Plot single_temperature_3mm_6mm/Cap_Par_Extraction/cgs_vds_ciss/ciss



Plot single_temperature_3mm_6mm/Cap_Par_Extraction/cgd_vds_crss/crss



Plot single_temperature_3mm_6mm/Cap_Par_Extraction/cds_vds_coss/coss



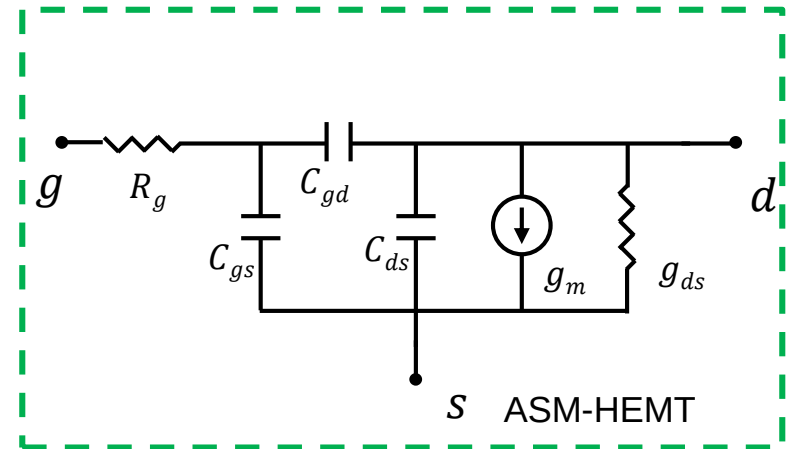
ASM HEMT - Small Signal Representation

$$Y_{11} = \frac{j\omega(C_{gs} + C_{gd})}{1 + \omega^2(C_{gs} + C_{gd})^2 R_g^2} + \frac{\omega^2(C_{gs} + C_{gd})^2 R_g}{1 + \omega^2(C_{gs} + C_{gd})^2 R_g^2}$$

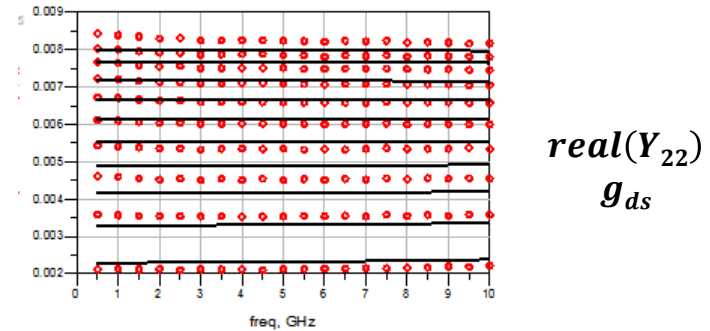
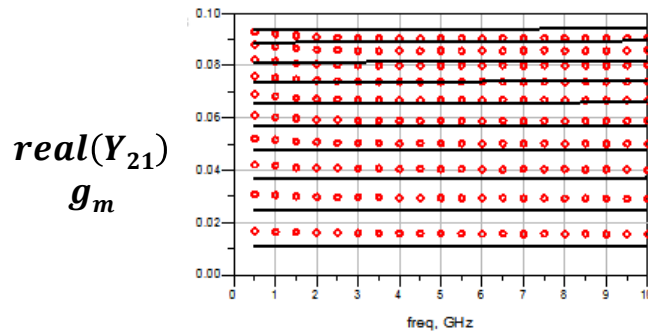
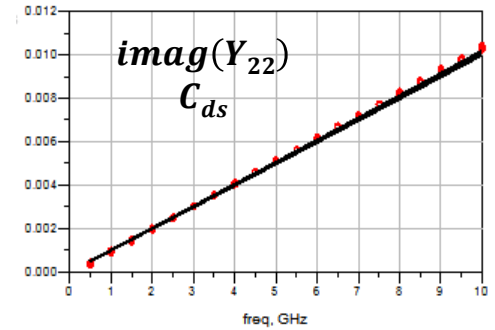
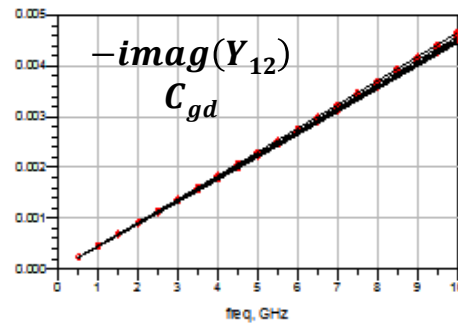
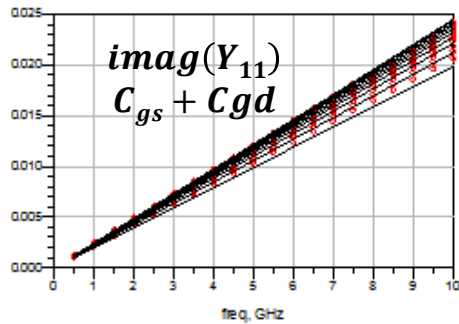
$$Y_{12} = -j\omega C_{gd} - \omega^2 C_{gd}(C_{gs} + C_{gd})R_g$$

$$Y_{21} = [g_m - \omega^2 C_{gd}(C_{gs} + C_{gd})R_g] - j\omega\{C_{gd} + g_m(C_{gs} + C_{gd})R_g\}$$

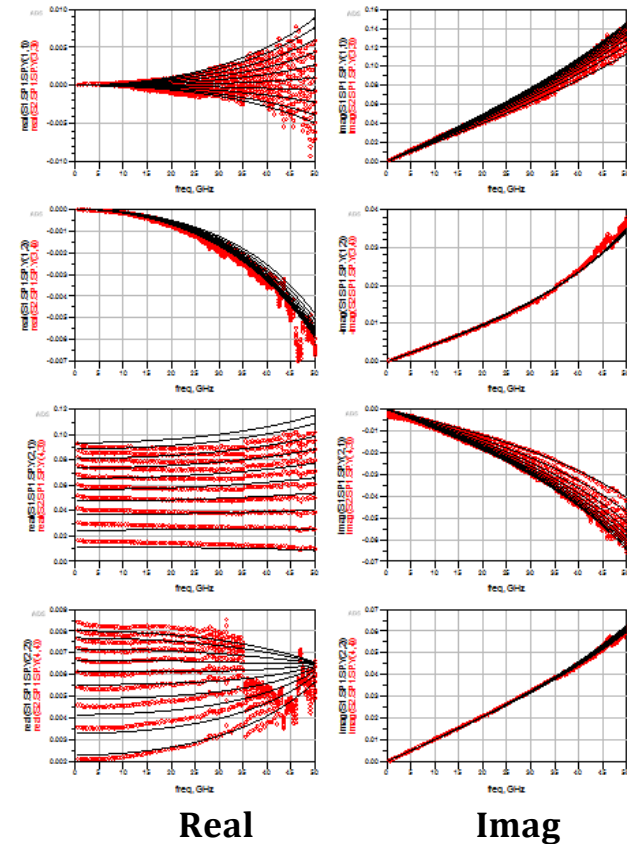
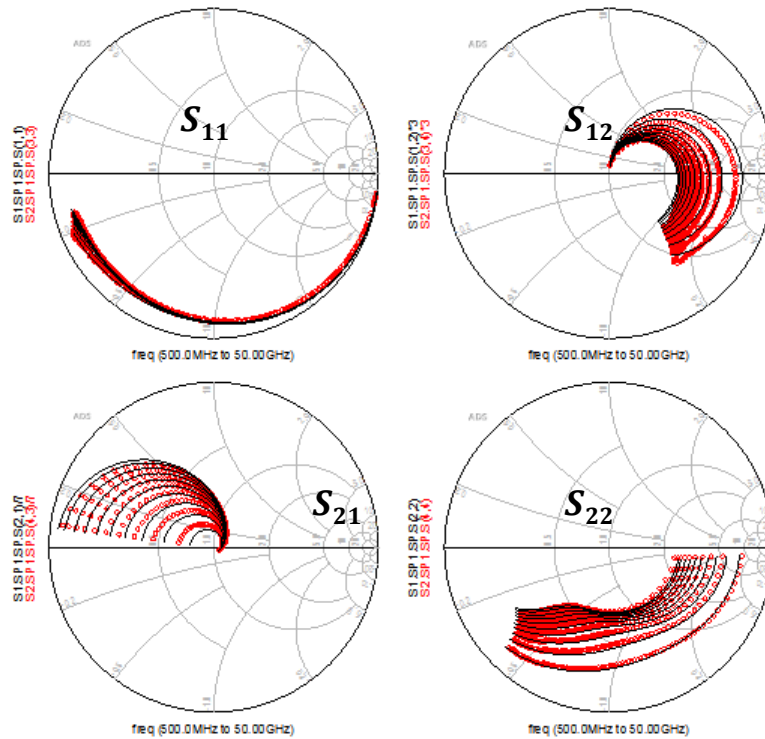
$$Y_{22} = g_{ds} + j\omega\{C_{ds} + C_{gd}(1 + g_m R_g)\}$$



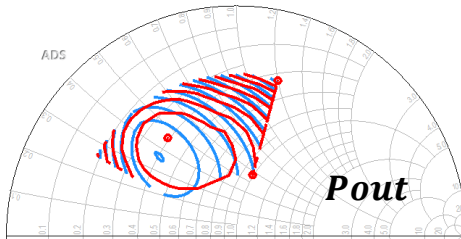
De-embedded Y-Parameters (0.5-10 GHz)



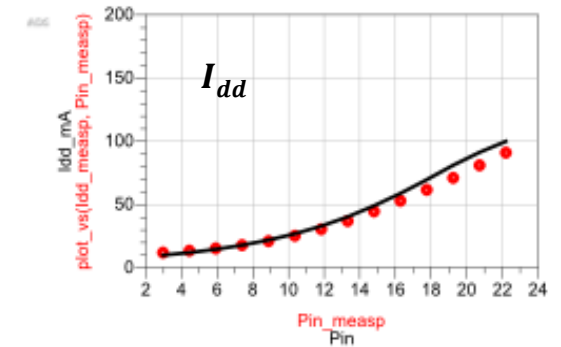
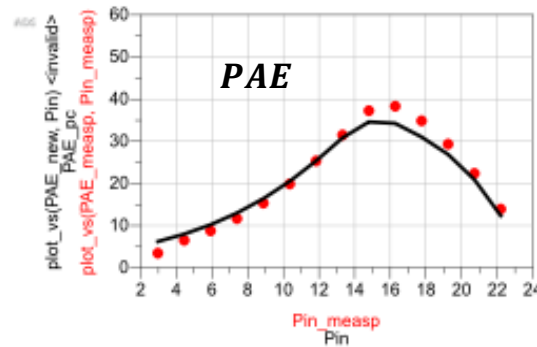
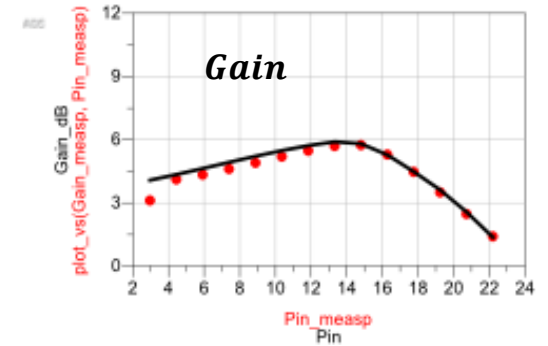
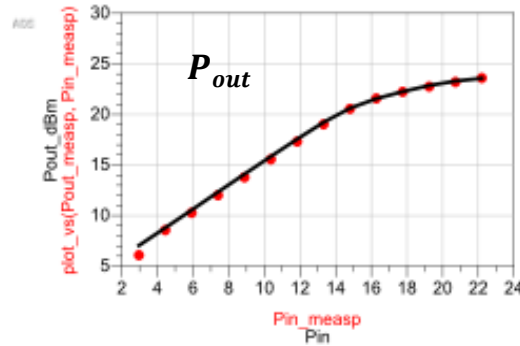
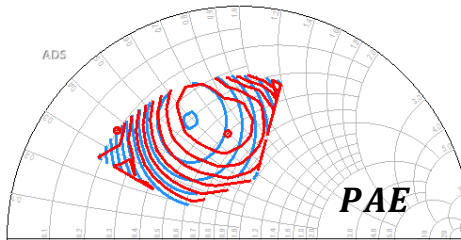
Broadband S-Parameters (0.5-50 GHz)



Load-Pull and Harmonic Balance Simulations



— Measured — Model



Agenda

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- **Summary**

Summary

- As one of the candidates for CMC GaN HEMT, ASM-HEMT is a physical model base on surface potential analytical calculation
- Model extraction procedure
 - DC parameter extraction – Focused on key parameters
 - Model field plate as a transistor and perform capacitance parameter extraction
 - RF modeling
 - Broadband S-Parameters and Y-Parameters
 - Large signal load pull and harmonic balance power sweeps
- The model is developed by using IC-CAP and ADS. Good fitting is achieved with industry measured data.

Acknowledgement and References

– For the helpful discussion and contribution

- Sheikh Aamir Ahsan, Sudip Ghosh, and Yogesh Singh Chauhan (IIT Kanpur)
- Sourabh Khandelwal (UC Berkeley)

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